

Scalability & Future Plan

Date	15 March 2024
Team ID	6ab4c24ba238dc999eb5437a
Project Name	AI BlogNest API
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single-server deployment on one Node.js/Express instance.	Higher traffic can overload the application instance.	HIGH
2	MongoDB/Mongoose is used for persistent blog and user data; no scaling strategy is documented.	Large data volume may require indexing, replication, and partitioning strategies.	MEDIUM
3	Basic Express middleware and synchronous request flow are documented; no caching or queueing layer is shown.	AI/API calls and database operations may increase response time under load.	HIGH

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Node.js/Express application instance on port 5000.	Use multiple application instances behind a load balancer and scale horizontally.
2	Data Storage	MongoDB accessed through Mongoose.	Add indexes, MongoDB replication and appropriate backup/retention strategy as data volume grows.

3	Performance	Express JSON parsing, CORS and Morgan logging; AI features depend on Gemini API.	Add caching for repeated reads, request queues for heavy AI work, monitoring and performance profiling.
4	Security	JWT authentication, bcryptjs, environment variables, CORS and express-validator dependency are included.	Add rate limiting, stronger secret/key management, HTTPS, security headers and regular dependency updates.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Add API rate limiting, security headers, monitoring and database indexes.	1–2 months	Improves security, observability and database response time.
Phase 2	Introduce horizontal application scaling, caching and background processing for AI requests.	3–6 months	Supports higher concurrent traffic and reduces request latency.
Phase 3	Add advanced analytics, multilingual AI generation, richer content workflows and production-grade cloud deployment.	6–12 months	Expands functionality and supports larger user and content volumes.